AM5920 - Deep learning for Mechanics

Assignment No. 1, Due Date: 26 Feb, 2026

1. Choose your favorite function $\mathbf{y} = \mathbf{f}(\mathbf{x})$, sample M data points from it in a domain and add some noise. This becomes your input paired dataset $(\{\mathbf{x}_i, \mathbf{y}_i\})$, $1 \leq i \leq M$. Use this dataset to construct a linear regression model. Investigate the convergence of your regression. Justify your choice of parameters in the algorithm, such as depth, width of the network.

2. For the function chosen above and using the dataset same as above, construct a simple deep neural network for the supervised learning/regression. Investigate the convergence of your algorithm. Justify your choice of parameters in the algorithm, such as depth, width of the network.

3. For the function chosen above and using the dataset same as above, construct a ResNet. Investigate the convergence of your algorithm. Justify your choice of parameters in the algorithm, such as depth, width of the network. Comment on generalisability of the network.